

# APPENDIX A

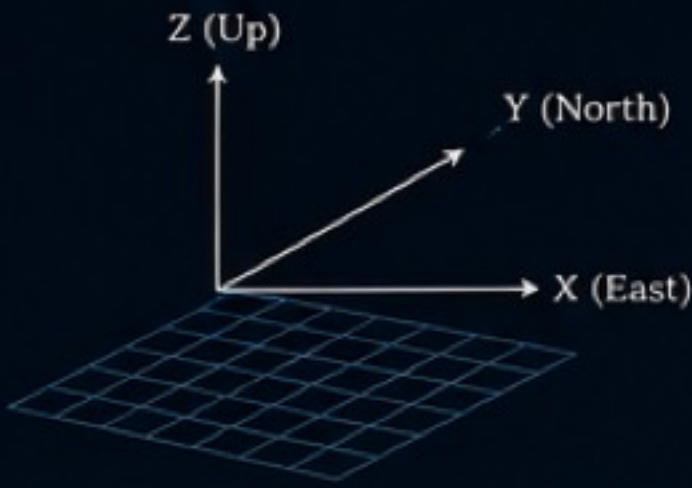
# AEROSPACE MATHEMATICS

## MODELS, EQUATIONS, AND ALGORITHMS

This appendix documents the mathematical foundations used in the Lunar Autonomous Landing Zone Assessment application.

### 1 COORDINATE SYSTEM

A local Cartesian frame is used to model the terrain.  
Origin at map center.



$x, y \in [-500, 500]$  m  
 $z$  : Elevation (m)

Grid spacing (uniform):  
 $\Delta = \frac{1000}{N-1}$  meters

### 2 TERRAIN MODEL GENERATION

The synthetic lunar terrain elevation  $z(x, y)$  is modeled as:

$$z(x, y) = \sum_i A_i \sin\left(\frac{x}{\lambda_{x,i}}\right) + \sum_j B_j \cos\left(\frac{y}{\lambda_{y,j}}\right) + \sum_k C_k \sin\left(\frac{x+y}{\lambda_{d,k}}\right) + \epsilon(x, y) + Z_{\text{craters}} + Z_{\text{boulders}}$$

$A_i, B_j, C_k$  : Amplitudes (m)

$\lambda_k$  : Spatial wavelengths (m)

$\epsilon(x, y)$  : Zero-mean Gaussian noise  $\sim \mathcal{N}(0, \sigma^2)$

$Z_{\text{craters}}, Z_{\text{boulders}}$  : Localized features (see below)



### 3 CRATER & BOULDER MODELS

Crater (negative Gaussian bowl with rim):

$$z_{\text{crater}}(r) = -d e^{-\left(\frac{r}{R}\right)^2} + 0.33d e^{-\left(\frac{r-R}{0.22R}\right)^2}$$

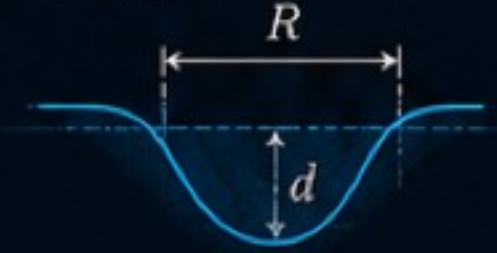
where:

$$r = \sqrt{(x - x_c)^2 + (y - y_c)^2}$$

$R$  : Crater radius (m)

$d$  : Crater depth (m)

$(x_c, y_c)$  : Crater center (m)



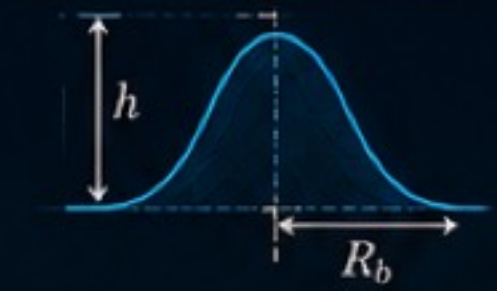
Boulder (positive Gaussian hill):

$$z_{\text{boulder}}(r) = h e^{-\left(\frac{r}{R_b}\right)^2}$$

where:

$R_b$  : Boulder radius (m)

$h$  : Boulder height (m)



### 4 SLOPE CALCULATION

Terrain slope at each grid cell is computed from the gradient of the elevation field.

$$\frac{\partial z}{\partial x} \approx \frac{z_{i,j+1} - z_{i,j-1}}{2\Delta}$$

$$\frac{\partial z}{\partial y} \approx \frac{z_{i+1,j} - z_{i-1,j}}{2\Delta}$$

$$\theta = \tan^{-1} \left( \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} \right)$$

$\theta$  : Slope angle (degrees)

$\Delta$  : Grid spacing (m)

#### SLOPE SCORE

$$S_{\text{slope}} = \text{clamp} \left( 1 - \frac{\theta}{18^\circ}, 0, 1 \right)$$

(Lower slope  $\rightarrow$  Higher score)

### 5 HAZARD PENALTY FIELDS

Hazards (craters and boulders) influence a penalty field that decreases with distance.

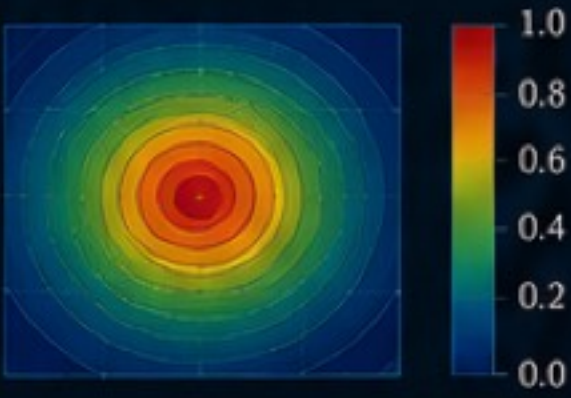
$$P(x, y) = \sum_k w_k e^{-\left(\frac{d_k(x, y)}{\sigma_k}\right)^2}$$

where:

$d_k(x, y)$  : Distance to feature  $k$

$w_k$  : Weight proportional to severity

$\sigma_k$  : Influence radius (m)



Hazard score (1 = safe, 0 = hazardous):

$$S_{\text{hazard}} = \text{clamp} \left( 1 - (0.58 P_{\text{crater}} + 0.42 P_{\text{boulder}}), 0, 1 \right)$$

### 6 ENVIRONMENTAL SCORE MODELS

#### Solar Exposure Score

$$S_{\text{solar}} = \text{clamp} \left( 0.45 + 0.55 Z_n + 0.08 \sin\left(\frac{x}{170}\right), 0, 1 \right)$$

$Z_n$  : Normalized elevation (0 to 1)



#### Communication Coverage Score

$$S_{\text{comm}} = \text{clamp} \left( 0.60 + 0.25 \cos\left(\frac{x-y}{260}\right) + 0.15 Z_n, 0, 1 \right)$$



### 7 OVERALL LANDING ZONE SUITABILITY SCORE

The final Landing Zone (LZ) score is a weighted combination of all factors.

$$S_{LZ} = 100 \times \frac{w_h S_{\text{hazard}} + 0.30 S_{\text{slope}} + w_s S_{\text{solar}} + w_c S_{\text{comm}}}{w_h + 0.30 + w_s + w_c}$$

$w_h$  : Hazard avoidance weight

$w_s$  : Solar exposure weight

$w_c$  : Communication coverage weight

#### RISK CLASSIFICATION

- LOW RISK  $S_{LZ} \geq 85$
- MODERATE RISK  $70 \leq S_{LZ} < 85$
- HIGH RISK  $S_{LZ} < 70$

### 8 ROVER PATH PLANNING

Rover survey waypoints are placed on a variable-radius star pattern about the LZ.

$$\theta_i = \frac{2\pi i}{N}, \quad i = 0, 1, \dots, N-1$$

$$r_i = R \left( 0.65 + 0.35 \times \frac{i \bmod 3}{2} \right)$$

$$\begin{aligned} x_i &= x_{LZ} + r_i \cos \theta_i \\ y_i &= y_{LZ} + r_i \sin \theta_i \end{aligned}$$



Path length:

$$L = \sum_{i=0}^{M-2} \sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} - y_i)^2}$$

Survey time:

$$t = \frac{L}{v_{\text{rover}}}, \quad v_{\text{rover}} : \text{Rover speed (m/s)}, \quad t : \text{Time (s)}$$

### 9 ADDITIONAL CONVERSIONS & CONSTANTS

#### Unit Conversions

1 km = 1000 m

1 hr = 3600 s

1 deg =  $\frac{\pi}{180}$  rad

#### Lunar Constants (mean values)

Surface gravity ( $g_{\text{Moon}}$ ) = 1.622 m/s<sup>2</sup>

Radius ( $R_{\text{Moon}}$ ) = 1,737.4 km

Day length = 29.53 Earth days

Escape velocity ( $v_{\text{esc}}$ )  $\approx$  2.38 km/s

#### Assumption Note

This application uses simplified models and surrogate metrics for concept development and decision support. Use mission-specific data, validated tools, and detailed analysis for flight-critical planning.

